

Answer Key — Electricity: Magnetic and Heating Effects

Final answers with brief reasons.

Objective

- Q1 **(b) magnetic** — current produces a magnetic field — the magnetic effect.
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- Q2 **(b) increase the number of turns** — more turns (and more current, and an iron core) make it stronger.
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- Q3 **(a) has high resistance and gives more heat** — high resistance produces more heat for a given current.
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- Q4 **(b) chemical reactions inside it** — a chemical reaction inside the cell makes the current.
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- Q5 **(b) carbon rod** — the central carbon rod is positive; the zinc container is negative.
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- Q6 **(a) True** — Oersted, in 1820.
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- Q7 **electrolyte** — the conducting liquid is the electrolyte.
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- Q8 **electromagnet** — it is an electromagnet.
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Short answer

- Q9 Increase the current (more cells) and increase the number of turns (or add an iron core).
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- Q10 Copper has low resistance (carries current with little heating); nichrome has high resistance (produces a lot of heat).
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- Q11 (i) Electric iron — heating effect; (ii) electric bell — magnetic effect.
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Long / application

- Q12 Wind insulated wire on an iron nail, connect to a cell; the nail attracts iron clips and deflects a compass when current flows, showing a magnetic field. Stronger by: more current (more cells), more turns, and using an iron core.
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- Q13 **Heating effect:** the wire's resistance converts electrical energy into heat, warming it. **Magnetic effect:** the current produces a magnetic field that deflects the compass needle. One current can show both effects.
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